

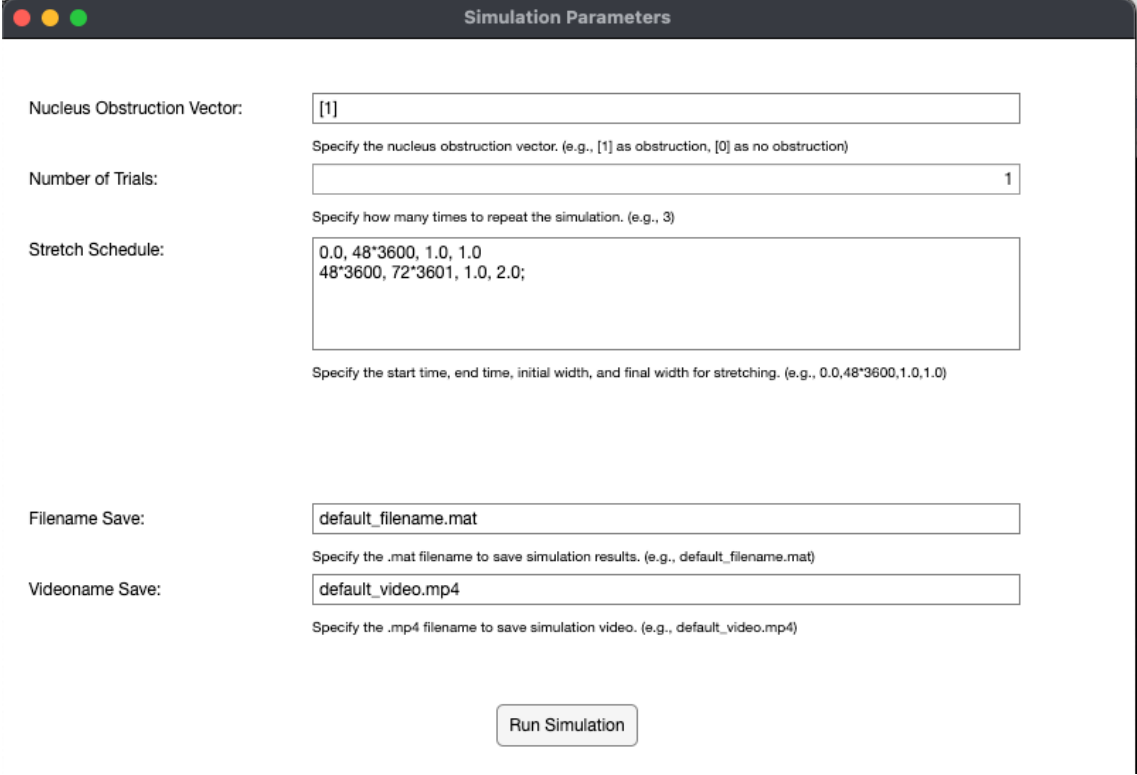
README

How to run the dynamic geometry change simulation.

First, install MATLAB and run `simulation_gui_dynamic_geometry.m` in singlecell folder.

Note that you need to choose branch 'feature/takashi/stretching' to use the dynamic geometry change feature.

Then you will see this interface pops up.



Simulation Parameters

Nucleus Obstruction Vector:
Specify the nucleus obstruction vector. (e.g., [1] as obstruction, [0] as no obstruction)

Number of Trials:
Specify how many times to repeat the simulation. (e.g., 3)

Stretch Schedule:
Specify the start time, end time, initial width, and final width for stretching. (e.g., 0.0, 48*3600, 1.0, 1.0)

Filename Save:
Specify the .mat filename to save simulation results. (e.g., default_filename.mat)

Videoname Save:
Specify the .mp4 filename to save simulation video. (e.g., default_video.mp4)

Nucleus Obstruction Vector

Specify the nucleus obstruction vector. Use [1] to treat the nucleus as an obstruction or [0] if there is no obstruction.

Example: [1].

Number of Trials

Define the number of times the simulation should repeat.

Example: 3.

Stretch Schedule

Specify the start time, end time, initial width, and final width for cell stretching.

Example:

0.0, 48*3600, 1.0, 1.0;

48*3600, 72*3601, 1.0, 2.0;

Filename Save

Specify the filename for saving the simulation results in .mat format.

Example: default_filename.mat.

Videoname Save

Specify the filename for saving the simulation video in .mp4 format.

Example: default_video.mp4